



DSTL (BCS-303) — Unit 4

Previous Year Questions with Step-by-Step Solutions

ABES Engineering College

Prepared for Practice and Revision

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How to Use

Each PYQ is followed by a line-by-line solution. For Hasse diagrams, we explain the construction steps (levels/covers). Redraw neatly in your answer booklet.

1 III Semester Theory Examination 2018–19

Q1(b) – Definition of Ring and Field

Define Ring and Field.

Solution

Ring. A nonempty set R together with two binary operations $+$ and $\cdot$ is called a *ring* if:

(R1) $(R, +)$ is an abelian (commutative) group.

- $a + b \in R$ for all $a, b \in R$ (closure).
- $a + (b + c) = (a + b) + c$ (associativity).
- There exists $0 \in R$ such that $a + 0 = 0 + a = a$ (additive identity).
- For each $a \in R$ there exists $-a \in R$ such that $a + (-a) = 0$ (additive inverse).
- $a + b = b + a$ (commutativity).

(R2) Multiplication is associative: $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all $a, b, c \in R$.

(R3) Multiplication is closed on R .

(R4) Distributive laws hold:

$$a \cdot (b + c) = a \cdot b + a \cdot c, \quad (a + b) \cdot c = a \cdot c + b \cdot c.$$

If there exists a multiplicative identity 1_R such that $1_R a = a 1_R = a$ for all $a \in R$, the ring is called a *ring with unity*. If multiplication is also commutative, $a \cdot b = b \cdot a$, the ring is called a *commutative ring*.

Field. A field F is a commutative ring with unity $1 \neq 0$ such that every nonzero element has a multiplicative inverse. Equivalently:

- $(F, +)$ is an abelian group (additive group).
- $(F \setminus \{0\}, \cdot)$ is an abelian group (multiplicative group).
- Multiplication distributes over addition.

Examples.

- $(\mathbb{Z}, +, \cdot)$ is a commutative ring with unity but not a field (integers $\neq 0$ do not all have multiplicative inverses in $\mathbb{Z}$).
- $(\mathbb{Q}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$, $(\mathbb{C}, +, \cdot)$ are fields.

Q2(b)(i) – Order of Subgroup Divides Order of Group

Let H be a subgroup of a finite group G . Prove that the order of H is a divisor of the order of G .

Solution

This is exactly **Lagrange's Theorem**.

Idea. Partition G into disjoint left cosets of H . Each coset has the same number of elements as H .

Proof. Let $|G| = n$ and $|H| = m$. Consider the set of left cosets of H in G :

$$G = a_1H \dot{\cup} a_2H \dot{\cup} \cdots \dot{\cup} a_kH,$$

where the union is disjoint and each a_iH is a distinct coset.

Define $f : H \rightarrow aH$ by $f(h) = ah$. This map is bijective (its inverse is $f^{-1}(ah) = h$), so $|aH| = |H| = m$ for every coset aH .

Since G is the disjoint union of k cosets, we have

$$|G| = k \cdot |H| = km.$$

Hence $m \mid n$, i.e. the order of H divides the order of G .

Q2(b)(ii) – Group of Prime Order is Cyclic

Prove that every group of prime order is cyclic.

Solution

Let G be a group with $|G| = p$, where p is prime.

Take any element $g \in G$, $g \neq e$. Consider the subgroup $\langle g \rangle$ generated by g . By Lagrange's theorem, the order $|\langle g \rangle|$ must divide $|G| = p$. Since p is prime and $|\langle g \rangle| \neq 1$ (because $g \neq e$), the only possibility is

$$|\langle g \rangle| = p.$$

Thus $\langle g \rangle$ has the same order as G , hence $\langle g \rangle = G$. So G is generated by g , and therefore G is a **cyclic group**.

Q4(a) – Algebraic Structure on $\{1, -1, i, -i\}$

Let $G = \{1, -1, i, -i\}$ with the binary operation multiplication, where $i = \sqrt{-1}$. Determine whether G is an algebraic structure and whether G is an Abelian group or not.

Solution

We interpret “algebraic structure” here as “group under multiplication”.

Closure. The product of any two elements is again in G :

$$i^2 = -1, \quad (-i)^2 = -1, \quad i \cdot (-i) = -i^2 = 1, \quad 1 \cdot x = x, \quad (-1) \cdot x = -x, \text{ etc.}$$

Cayley table of $(G, \cdot)$:

$\cdot$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

From the table:

- 1 acts as identity: $1x = x1 = x$.

- Each element has an inverse:

$$1^{-1} = 1, \quad (-1)^{-1} = -1, \quad i^{-1} = -i, \quad (-i)^{-1} = i.$$

- Multiplication is associative and commutative (inherited from complex-number multiplication); the table is symmetric about the main diagonal, so $ab = ba$.

Hence $(G, \cdot)$ is a **group**, and because multiplication is commutative, it is an **abelian group**.

Q4(b) – Meaning of Ring; Examples

What is meant by a ring? Give examples of both commutative and non-commutative rings.

Solution

A ring was defined in Q1(b). Here we emphasize the examples.

Commutative ring examples.

- $(\mathbb{Z}, +, \cdot)$ — integers with usual addition and multiplication.
- $(\mathbb{Z}_n, +, \cdot)$ — integers modulo n with addition and multiplication modulo n .
- Polynomial ring $R[x]$ over a commutative ring R .

Non-commutative ring examples.

- The ring $M_n(\mathbb{R})$ of all $n \times n$ real matrices: matrix multiplication is not commutative when $n \geq 2$.
- The ring of 2×2 complex matrices $M_2(\mathbb{C})$.

2 III Semester Theory Examination 2019–20

Q1(c) – Identity Element in $(\mathbb{Z}, *)$

Let $\mathbb{Z}$ be the group of integers with binary operation $*$ defined by

$$a * b = a + b - 2, \quad \text{for all } a, b \in \mathbb{Z}.$$

Find the identity element of the group $(\mathbb{Z}, *)$.

Solution

We seek $e \in \mathbb{Z}$ such that for all $a \in \mathbb{Z}$,

$$a * e = a \quad \text{and} \quad e * a = a.$$

Using the definition:

$$a * e = a + e - 2 = a \quad \Rightarrow \quad e - 2 = 0 \Rightarrow e = 2.$$

Similarly,

$$e * a = e + a - 2 = a \quad \Rightarrow \quad e - 2 = 0 \Rightarrow e = 2.$$

So $e = 2$ is the identity element in $(\mathbb{Z}, *)$.

Q1(d) – Every Cyclic Group is Abelian

Show that every cyclic group is abelian.

Solution

Let G be a cyclic group generated by g , i.e. $G = \langle g \rangle$. Every element of G can be written as a power of g :

$$\forall a \in G, \exists m \in \mathbb{Z} \text{ such that } a = g^m.$$

Take arbitrary elements $a = g^m$ and $b = g^n$ in G . Using associativity of the group operation (written multiplicatively),

$$ab = g^m g^n = g^{m+n} = g^{n+m} = g^n g^m = ba.$$

Thus $ab = ba$ for all $a, b \in G$. So G is an abelian group.

Q2(b) – Cosets of a Subgroup in $\mathbb{Z}$

What do you mean by cosets of a subgroup? Consider the group $\mathbb{Z}$ of integers under addition and the subgroup

$$H = \{\dots, -12, -6, 0, 6, 12, \dots\}$$

consisting of multiples of 6.

- (i) Find the cosets of H in $\mathbb{Z}$.
- (ii) What is the index of H in $\mathbb{Z}$?

Solution

Coset definition. If $(G, +)$ is a group and $H \leq G$ a subgroup, then for any $a \in G$, the *left coset* of H containing a is

$$a + H = \{a + h \mid h \in H\}.$$

In an abelian group, left and right cosets coincide.

Here $G = \mathbb{Z}$, $H = 6\mathbb{Z}$. Cosets are of the form $a + 6\mathbb{Z}$.

Any integer n is congruent to exactly one of $0, 1, 2, 3, 4, 5 \pmod{6}$. Hence there are exactly six distinct cosets:

$$0 + H = \{\dots, -12, -6, 0, 6, 12, \dots\},$$

$$1 + H = \{\dots, -11, -5, 1, 7, 13, \dots\},$$

$$2 + H = \{\dots, -10, -4, 2, 8, 14, \dots\},$$

$$3 + H = \{\dots, -9, -3, 3, 9, 15, \dots\},$$

$$4 + H = \{\dots, -8, -2, 4, 10, 16, \dots\},$$

$$5 + H = \{\dots, -7, -1, 5, 11, 17, \dots\}.$$

Thus there are 6 cosets; the **index** of H in G is

$$[\mathbb{Z} : H] = 6.$$

Q4(a) – Ring and Its Elementary Properties

What is Ring? Define elementary properties of Ring with example.

Solution

A ring was defined earlier. Its elementary properties summarize as:

1. $(R, +)$ is an abelian group.
2. Multiplication is associative and closed.
3. Distributive laws link $+$ and $\cdot$.
4. A ring may or may not have multiplicative identity; multiplication may or may not be commutative.

Example. $(\mathbb{Z}, +, \cdot)$ satisfies all ring axioms:

- Integers form an abelian group under addition.
- Integer multiplication is associative and distributive over addition.
- 1 acts as multiplicative identity.

Hence $\mathbb{Z}$ is a (commutative) ring with unity.

Q4(b) – Intersection of Normal Subgroups

Prove or disprove that the intersection of two normal subgroups of a group G is again a normal subgroup of G .

Solution

Let N_1, N_2 be normal subgroups of G . Consider $N = N_1 \cap N_2$.

(i) N is a subgroup.

- Nonempty: $e \in N_1, N_2 \Rightarrow e \in N$.
- If $x, y \in N$, then $x, y \in N_1$ and $x, y \in N_2$, so $xy^{-1} \in N_1$ and $xy^{-1} \in N_2$; hence $xy^{-1} \in N$.

Thus $N \leq G$.

(ii) N is normal. Take any $g \in G$ and $n \in N$. Then $n \in N_1$ and $n \in N_2$. Since N_1, N_2 are normal,

$$gng^{-1} \in N_1, \quad gng^{-1} \in N_2.$$

So $gng^{-1} \in N_1 \cap N_2 = N$. Hence $gNg^{-1} \subseteq N$. Similarly, for any $m \in N$ there exists $n = g^{-1}mg \in N$, so $m = gng^{-1} \in gNg^{-1}$ and we get $N \subseteq gNg^{-1}$. Therefore $gNg^{-1} = N$ for all $g \in G$. So N is normal.

Thus the intersection of two normal subgroups is again a **normal subgroup**.

3 III Semester Theory Examination 2020–21

Q1(c) – Define Group

Define group.

Solution

A **group** $(G, *)$ is a nonempty set G with a binary operation $*$ satisfying:

1. **Closure:** $a * b \in G$ for all $a, b \in G$.
2. **Associativity:** $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$.
3. **Identity:** There exists $e \in G$ such that $e * a = a * e = a$ for all $a \in G$.
4. **Inverse:** For each $a \in G$ there exists $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

If additionally $a * b = b * a$ for all $a, b \in G$, then G is an **abelian group**.

Q1(d) – Define Ring

Define ring.

Solution

Same as earlier definition; we just recall: a ring $(R, +, \cdot)$ is a set closed under two operations such that $(R, +)$ is an abelian group, $\cdot$ is associative and distributive over $+$.

Q2(b) – Lagrange’s Theorem for Groups

State and prove Lagrange theorem for group.

Solution

Statement. If G is a finite group and $H \leq G$, then $|H|$ divides $|G|$. Moreover,

$$|G| = [G : H] \cdot |H|,$$

where $[G : H]$ is the number of (left) cosets of H in G .

Proof. Same as in Q2(b)(i) of 2018–19 and Q2(b) of 2019–20: partition G into disjoint cosets of H , each coset has size $|H|$, so $|G|$ is a multiple of $|H|$.

Q4(a) – Characterization of Normal Subgroup

A subgroup H of a group G is a normal subgroup if and only if $g^{-1}hg \in H$ for every $h \in H$ and $g \in G$. Justify.

Solution

By definition, $H \trianglelefteq G$ (H normal) means

$$gH = Hg \quad \text{for all } g \in G.$$

($\Rightarrow$) Assume H is normal. Then for each $g \in G$,

$$gH = Hg.$$

Take any $h \in H$. Because $gh \in gH$, there exists $h' \in H$ with $gh = h'g$. Rearranging:

$$g^{-1}h'g = h \in H.$$

Equivalently, $g^{-1}hg \in H$. So the conjugate of any $h \in H$ by any $g \in G$ lies in H .

($\Leftarrow$) Suppose $g^{-1}hg \in H$ for all $h \in H$ and $g \in G$. Take any $g \in G$. For each $h \in H$,

$$gh = (ghg^{-1})g,$$

where $ghg^{-1} \in H$ by assumption. Thus $gh \in Hg$, so $gH \subseteq Hg$. The reverse inclusion is similar (multiply on left by g^{-1}). Therefore $gH = Hg$. Hence H is normal.

Q4(b) – Basic Identities in a Group

In a group $(G, *)$ prove that

(i) $(a^{-1})^{-1} = a,$

(ii) $(ab)^{-1} = b^{-1}a^{-1}.$

Solution

(i) $(a^{-1})^{-1} = a$. By definition, a^{-1} is the unique element such that $a * a^{-1} = a^{-1} * a = e$. Now a is an inverse of a^{-1} as well, so by uniqueness of inverses, $(a^{-1})^{-1} = a$.

(ii) $(ab)^{-1} = b^{-1}a^{-1}$. Check:

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = aea^{-1} = aa^{-1} = e,$$

and similarly,

$$(b^{-1}a^{-1})(ab) = b^{-1}(a^{-1}a)b = b^{-1}eb = bb^{-1} = e.$$

Thus $b^{-1}a^{-1}$ is both a left and right inverse of ab . By uniqueness, $(ab)^{-1} = b^{-1}a^{-1}$.

4 III Semester Theory Examination 2021–22

Q1(c) – Every Cyclic Group is Abelian

State and justify “Every cyclic group is an abelian group”.

Solution

Statement and proof are identical to Q1(d) of 2019–20: if $G = \langle g \rangle$ then every element is g^m . For $a = g^m, b = g^n, ab = g^{m+n} = g^{n+m} = ba$. Hence cyclic $\Rightarrow$ abelian.

Q1(d) – State Ring and Field with Example

State Ring and Field with example.

Solution

Ring and field definitions as in Q1(b) of 2018–19 with examples like $\mathbb{Z}$ (ring) and $\mathbb{Q}$ (field).

Q2(b) – Cyclic Group and Order of Subgroup

Explain cyclic group. Let H be a subgroup of a finite group G . Justify the statement “the order of H is a divisor of the order of G ”.

Solution

A **cyclic group** is a group generated by a single element $g \in G$; we write $G = \langle g \rangle$. The second part is simply Lagrange's theorem again: see earlier proof.

Q4(a) – $(\mathbb{Z}, *)$ with $x * y = x + y + 1$

Define the binary operation $*$ on $\mathbb{Z}$ by

$$x * y = x + y + 1 \quad \text{for all } x, y \in \mathbb{Z}.$$

Verify that $(\mathbb{Z}, *)$ is an abelian group. Discuss the properties of abelian group.

Solution

We check group axioms.

Closure. For integers x, y , $x + y + 1$ is an integer.

Associativity.

$$\begin{aligned} (x * y) * z &= (x + y + 1) * z = (x + y + 1) + z + 1 = x + y + z + 2, \\ x * (y * z) &= x * (y + z + 1) = x + (y + z + 1) + 1 = x + y + z + 2. \end{aligned}$$

Thus $(x * y) * z = x * (y * z)$.

Identity. Suppose e is identity: $x * e = x$ and $e * x = x$. Then

$$x * e = x + e + 1 = x \Rightarrow e + 1 = 0 \Rightarrow e = -1.$$

Similarly $e * x = e + x + 1 = x \Rightarrow e = -1$. So $e = -1$ is identity.

Inverse. For each $x \in \mathbb{Z}$, we want y such that $x * y = e = -1$:

$$x * y = x + y + 1 = -1 \Rightarrow y = -1 - x - 1 = -x - 2.$$

So the inverse of x is $-x - 2$, which is again an integer.

Commutativity.

$$x * y = x + y + 1 = y + x + 1 = y * x.$$

Hence $(\mathbb{Z}, *)$ is an abelian group.

Properties of abelian group. In any abelian group:

- The operation is commutative.
- Subgroups generated by any single element are cyclic.
- All left cosets equal right cosets.

Q4(b) – Intersection of Subgroups; Square of Product

- Justify that “The intersection of any two subgroups of a group $(G, *)$ is again a subgroup of $(G, *)$ ”.
- Justify that “If a, b are arbitrary elements of a group G then $(ab)^2 = a^2b^2$ if and only if G is abelian”.

Solution

(i) **Intersection of subgroups.** Proof is identical to part of Q4(b) 2019–20: for subgroups $H_1, H_2 \leq G$, their intersection is nonempty and closed under the operation and inverses, hence a subgroup.

(ii) **Characterization via $(ab)^2$.**

($\Rightarrow$) Assume G is abelian. Then $ab = ba$, so

$$(ab)^2 = (ab)(ab) = a(ba)b = a(ab)b = a^2b^2.$$

($\Leftarrow$) Assume $(ab)^2 = a^2b^2$ for all $a, b \in G$. Compute:

$$(ab)^2 = abab, \quad a^2b^2 = aabb.$$

So $abab = aabb$. Multiply on the left by a^{-1} and on the right by b^{-1} :

$$a^{-1}(abab)b^{-1} = a^{-1}(aabb)b^{-1} \Rightarrow bab = abb \Rightarrow ba = ab.$$

Thus $ab = ba$ for all $a, b \in G$; hence G is abelian.

5 III Semester Theory Examination 2022–23

Q1(i) – Define Normal Subgroup

Define normal subgroup.

Solution

A subgroup N of a group G is called a **normal subgroup** (denoted $N \trianglelefteq G$) if

$$gN = Ng \quad \text{for all } g \in G,$$

or equivalently if

$$g^{-1}ng \in N \quad \text{for all } n \in N, g \in G.$$

Q3(a) – Detailed Study of $G = \{1, -1, i, -i\}$

Let $G = \{1, -1, i, -i\}$ with the operation of ordinary multiplication on G , where $i = \sqrt{-1}$.

- (i) Determine whether G is abelian.
- (ii) Determine the order of each element in G .
- (iii) Determine whether G is a cyclic group; if G is a cyclic group then determine the generator/generators of the group G .
- (iv) Determine a subgroup of the group G .

Solution

We reuse the Cayley table from Q4(a) 2018–19:

$\cdot$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	1	-1
$-i$	$-i$	i	1	-1

(i) **Abelian?** The table is symmetric about the diagonal, so $ab = ba$ for all $a, b \in G$. Hence G is an **abelian group**.

(ii) **Order of elements.**

- $1^1 = 1 \Rightarrow \text{ord}(1) = 1$.
- $(-1)^2 = 1 \Rightarrow \text{ord}(-1) = 2$.
- $i^2 = -1, i^3 = -i, i^4 = 1 \Rightarrow \text{ord}(i) = 4$.
- Similarly, $(-i)^2 = -1, (-i)^3 = i, (-i)^4 = 1 \Rightarrow \text{ord}(-i) = 4$.

(iii) **Cyclic and generators.** Because $\text{ord}(i) = 4 = |G|$, the subgroup $\langle i \rangle$ has four elements, hence equals G . So G is cyclic with generator i . Also $-i = i^3$ is another generator; $\langle -i \rangle = G$ as well.

(iv) **A subgroup.** Several choices exist. For example,

$$H = \{1, -1\}$$

is a subgroup: closed under multiplication, contains identity, and each element has inverse within H ($1^{-1} = 1, (-1)^{-1} = -1$). It is of order 2.

Q3(b) – Basic Properties of Group Homomorphism

Let $(G, *)$ and $(G', *')$ be any two groups and let e and e' be their respective identities. If f is a homomorphism of G into G' , then prove that

- $f(e) = e'$,
- $f(x^{-1}) = [f(x)]^{-1}$ for all $x \in G$.

Solution

A homomorphism satisfies $f(x * y) = f(x) *' f(y)$.

(i) **Image of identity.** Since e is identity in G ,

$$f(e) = f(e * e) = f(e) *' f(e).$$

Let $f(e) = a \in G'$. Then $a = a *' a$. Multiply on the left by a^{-1} in G' :

$$e' = a^{-1} *' a = a^{-1} *' (a *' a) = (a^{-1} *' a) *' a = e' *' a = a.$$

Thus $a = e'$. So $f(e) = e'$.

(ii) **Image of inverse.** We know $x * x^{-1} = e$. Apply f :

$$f(x * x^{-1}) = f(e) \Rightarrow f(x) *' f(x^{-1}) = e'.$$

But e' is identity in G' . Thus $f(x^{-1})$ is a right inverse of $f(x)$. In a group, right inverse = left inverse = unique inverse. So $f(x^{-1}) = [f(x)]^{-1}$.

Q7(a) – Residue Classes Modulo 5 Form a Field

Prove that the set of residues $F = \{0, 1, 2, 3, 4\}$ modulo 5 is a field with respect to addition and multiplication of residue classes modulo 5, i.e. $(F, +_5, \times_5)$ is a field.

Solution

Here operations are defined modulo 5:

$$a +_5 b \equiv a + b \pmod{5}, \quad a \times_5 b \equiv ab \pmod{5}.$$

Additive structure. $(F, +_5)$ is an abelian group:

- Closed under $+_5$.
- Associativity and commutativity inherited from integers modulo 5.
- Additive identity is 0.
- Each element has additive inverse:

$$0^{-1} = 0, \quad 1^{-1} = 4, \quad 2^{-1} = 3, \quad 3^{-1} = 2, \quad 4^{-1} = 1 \pmod{5}.$$

Multiplicative structure on $F^\times = F \setminus \{0\} = \{1, 2, 3, 4\}$.

- Closed under $\times_5$.
- Associative and commutative (inherited).
- Identity is 1.
- Every nonzero element has a multiplicative inverse modulo 5:

$$1^{-1} = 1, \quad 2^{-1} = 3, \quad 3^{-1} = 2, \quad 4^{-1} = 4 \quad (\text{since } 2 \cdot 3 \equiv 6 \equiv 1, \quad 4 \cdot 4 \equiv 16 \equiv 1).$$

Thus $(F^\times, \times_5)$ is an abelian group.

Distributivity. Multiplication distributes over addition modulo 5 because this property holds in $\mathbb{Z}$ and remains valid under reduction modulo 5.

Hence F with these operations is a field: $\mathbb{Z}_5$.